

B28: SQL SuperHero Program : Session 23

1 message

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INSTR and SUBSTR

INSTR --> gives you a number as output , on which number position that character exists
SUBSTR --> gives you string as output .



The screenshot displays two SQL query results. The first query is `select instr('vishal', 'a') from dual`, which returns the value 5 in a table with the header `INSTR('VISHAL','A')`. Below the result is a `Download CSV` link. The second query is `select instr('vishal', 't') from dual`, which returns the value 0 in a table with the header `INSTR('VISHAL','T')`. Below this result is also a `Download CSV` link.

--> INSTR will give you > 0 number, if that character/string exists in the main string. Like in above example, 'VISHAL' is the string and we have to find out the position of 'A', so when we do `INSTR('Vishal','a')`, it gives 5, because it exists on 5th position.

In the above example, the output is 0, because 't' is not found in 'VISHAL'.

INSTR is a very useful and cost effective operation if we have to just check the data existence by using LIKE operation. we can avoid LIKE by using INSTR

```
select * from vishal123 where name like 'vis\%%' escape '\'
```

ID	NAME
3	vis%jais

```
select * from vishal123 where instr(name, 'vis%') > 0
```

ID	NAME
3	vis%jais

Both
same

column like '%vis%' = instr(column, 'vis') > 0

In this above example, it will just check if this particular column contains 'vis' as a value or not.

INSTR is a very powerful function and cost effective. We can easily replace LIKE by INSTR.

```
select instr('Abdul','b') from dual
```

INSTR('ABDUL','B')
2

→ checking the position of
'b' in 'Abdul'
→ its on 2nd position

```
select instr('Abdul','t') from dual
```

INSTR('ABDUL','T')
0

→ can not find 't' in
Abdul

ms (main string)

```
select instr('My name is khan','is') from dual
```

↳ what i want to search

INSTR('MYNAMEISKHAN','IS')

9

gives position

A	B	C	D	E	F	G	H	I	J	K	L	M	N	
O	R	A	C	L	E	D	A	T	A	B	A	S	E	
1	2	3	4	5	6	7	8	9	10	11	12	13	14	

In This word "ORACLEDATABASE", I have assigned a number to represent the position of each character.

Worksheet Query Builder

```
1 select INSTR('oracledatabase','a',1,2) FROM DUAL
```

Query Result x

SQL | All Rows Fetched: 1 in 0.523 seconds

INSTR('ORACLEDATABASE','A',1,2)
1 8

2nd Occurance of 'A'

By default INSTR will give you 1st occurrence of that string. If I have to search 2nd, 3rd, 4th we have to use the above syntax.

```
SELECT INSTR('ORACLEDATABASE','A',1,2) FROM DUAL
```

```
INSTR('ORACLEDATABASE','A',1,2)
```

8

2nd occurrence

```
SELECT INSTR('ORACLEDATABASE','A',1,3) FROM DUAL
```

```
INSTR('ORACLEDATABASE','A',1,3)
```

10

3rd

```
SELECT INSTR('ORACLEDATABASE','A',1,4) FROM DUAL
```

```
INSTR('ORACLEDATABASE','A',1,4)
```

12

4th

```
select INSTR('oracleatabase','a',-1) FROM DUAL;
```

minus means start counting from reverse. Here it is saying -1, so start from last and when it occurs 1st time, give that number.

Worksheet | Query Builder

```
1 select INSTR('oracleatabase','a',-1) FROM DUAL
```

Query Result x

SQL | All Rows Fetched: 1 in 0.52 seconds

	INSTR('ORACLEDATABASE','A',-1)
1	12

From Reverse 1st position

minus = Reverse

```
SELECT INSTR('ORACLEDATABASE','A',-1,2) FROM DUAL
```

```
INSTR('ORACLEDATABASE','A',-1,2)
```

10

2nd occurrence

```
SELECT INSTR('ORACLEDATABASE','A',-1,3) FROM DUAL
```

```
INSTR('ORACLEDATABASE','A',-1,3)
```

8

A	B	C	D	E	F	G	H	I	J	K	L	M	N
V	I	S	H	A	L		J	A	I	S	W	A	L
1	2	3	4	5	6	7	8	9	10	11	12	13	14

See the above image, we assigned a number to each letter.

```
SELECT INSTR('VISHAL JAISWAL','A',-1,1) FROM DUAL;
```

The above query try to search "A" in the main string "vishal jaiswal", 3rd parameter is -1 means reverse and last is 1 means 1st occurrence.

1 SELECT INSTR('VISHAL JAISWAL','A',-1,1) FROM DUAL

13th

INSTR('VISHALJAISWAL','A',-1,1)

13

Download CSV

Let's do a small twist and change the 3rd parameter :

1 SELECT INSTR('VISHAL JAISWAL','A',-4,1) FROM DUAL

minus = Reverse

- 4 means start after 4th position in reverse

INSTR('VISHALJAISWAL','A',-4,1)

9

Download CSV

SELECT INSTR('VISHAL JAISWAL','A',-4,1) FROM DUAL

here the 3rd parameter is -4. Minus means reverse and -4 means go to 4th position from reverse and then give me the first occurrence of "A"

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
	V	I	S	H	A	L		J	A	I	S	W	A	L
	1	2	3	4	5	6	7	8	9	10	11	12	13	14

select instr('GLORIA NTIRWIHORWA','I',1,2) from dual

from start
 2nd occurrence
 2nd occuran of 'i' from start

INSTR('GLORIANNTIRWIHORWA','I',1,2)

10

select instr('GLORIA NTIRWIHORWA','I',-1,1) from dual

from reverse 1st occurrence of 'i'

INSTR('GLORIANNTIRWIHORWA','I',-1,1)

13

select * from hr.employees where hire_date like '%JUN%'

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID
100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES
105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG
133	Jason	Mallin	JMALLIN	650.127.1934	14-JUN-04	ST_CLERK
182	Martha	Sullivan	MSULLIVA	650.507.9878	21-JUN-07	SH_CLERK

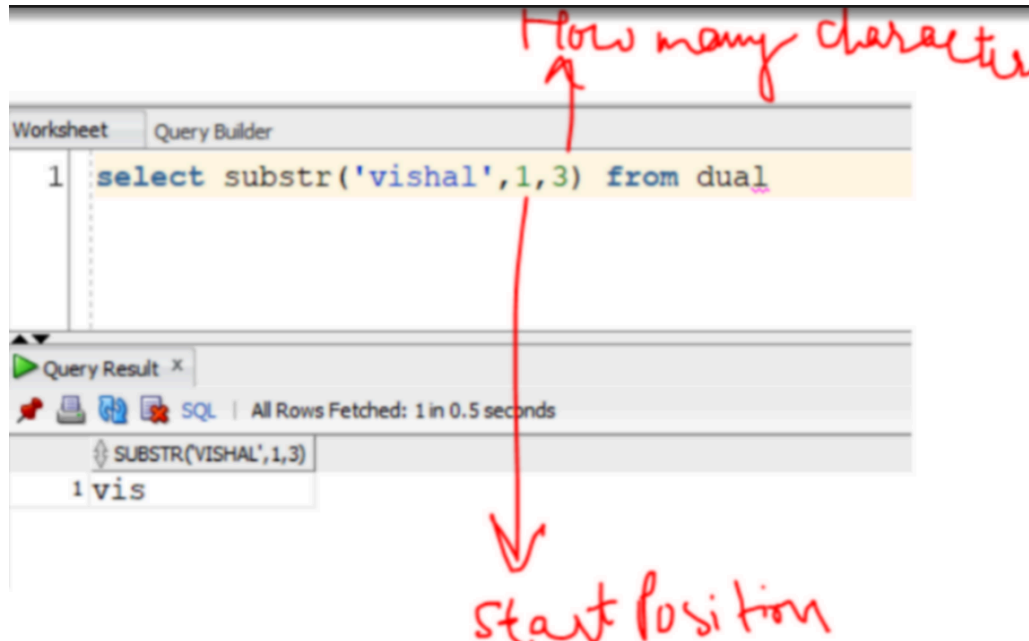
select * from hr.employees where instr(hire_date,'JUN') > 0

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID
100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES
105	David	Austin	DAUSTIN	590.423.4569	25-JUN-05	IT_PROG
133	Jason	Mallin	JMALLIN	650.127.1934	14-JUN-04	ST_CLERK
182	Martha	Sullivan	MSULLIVA	650.507.9878	21-JUN-07	SH_CLERK
186	Julia	Dellinger	JDELLING	650.509.3876	24-JUN-06	SH_CLERK

If you see the way of using INSTR or LIKE.

*****SUBSTR*****

Will give output as a string.



```
select substr('vishal',1,3) from dual;
```

--> This will give you "VIS", because start with 1st position and give us 3 more characters.



Again Minus is used in the same way. minus means reverse.

2 more characters

Worksheet Query Builder

```
1 select substr('akriti',-3,2) from dual
```

Query Result x

SQL | All Rows Fetched: 1 in 0.497 seconds

SUBSTR('AKRITI',-3,2)
iti

start from reverse because - sign go to 3rd last character to start

MS HML

```
select substr('kunal jain',-4,3) from dual
```

SUBSTR('KUNALJAIN',-4,3)
jai

SP → from reverse 4th position

Here the 2nd parameter is -3. It is minus, so it means string numbering in reverse order. So go to 3rd last character and give 2 more characters after that.

--> print last 5 character of word "oracledatabase"

```
select substr('oracledatabase',-5,5) from dual;
```

```
select substr('oracledatabase',-5,5) from dual
```

```
SUBSTR('ORACLEDATABASE',-5,5)
```

```
abase
```

start with last 5th because -5 is the 2nd parameter and give next 5 characters. so output is "abase"

--> Print last 4 character of your name?

```
SELECT substr('VISHAL JAISWAL',-4,4) FROM DUAL
```

```
1 SELECT substr('VISHAL JAISWAL',-4,4) FROM DUAL
```

```
SUBSTR('VISHALJAISWAL',-4,4)
```

```
SWAL
```

[Download CSV](#)

```
SELECT SUBSTR('ORACLEDATABASE',-5, 3) FROM DUAL
```

```
SUBSTR('ORACLEDATABASE',-5,3)
```

```
ABA
```

start position
11

From
Reverse
5th

SP

```
SELECT SUBSTR('SURAJ HIREKHAN', -4, 4) FROM DUAL
```

HMC

SUBSTR('SURAJHIREKHAN', -4, 4)
KHAN

HMC

```
SELECT SUBSTR('ORACLE', -2, 2) FROM DUAL
```

SP

SUBSTR('ORACLE', -2, 2)
LE

ms *SP* *HMC*

```
select substr('savita waghmare', 1, 5) from dual
```

*→ from 1st position
give me 5 more
character*

SUBSTR('SAVITAWAGHMARE', 1, 5)
savit

```
select substr('savita waghmare', -7, 4) from dual
```

*→ -7 → go to reverse 7th position
& give 4 more
character*

SUBSTR('SAVITAWAGHMARE', -7, 4)
aghm

ms / sp hmc

```
select substr('vishal jaiswal',7,3) from dual
```

→ from 7th position
3 more character

```
SUBSTR('VISHALJAISWAL',7,3)
```

```
ja
```

ms / sp hmc

```
select substr('vishal jaiswal',9,3) from dual
```

→ from 9th position
3 more character

```
SUBSTR('VISHALJAISWAL',9,3)
```

```
ais
```

ms / sp hmc

```
select substr('amrita waghmare',-3,7) from dual
```

```
SUBSTR('AMRITAWAGHMARE',-3,7)
```

```
are
```

start from Reverse -3
and 7 more characters
Do not have 7 then Print as
much as I have

ms

hmc

```
SELECT SUBSTR('SACHIN GODBOLE',2,100) FROM DUAL
```

```
SUBSTR('SACHINGODBOLE',2,100)
```

```
ACHIN GODBOLE
```

I don't have
100
so give
whatever
we
have

MS SP HMC

```
select substr('amrita waghmare',1,100) from dual
```

SUBSTR('AMRITAWAGHMARE',1,100)
amrita waghmare

I do not have 100 Characters, so if will give fill what he has

```
select substr('amrita waghmare',1) from dual
```

MS SP

SUBSTR('AMRITAWAGHMARE',1)
amrita waghmare

→ forgot HMC.
give every thing

Suppose we have to start from 'i' and pick next 6 string.
so if input is 'vishaljaiswal', so we have start from 'i' so we have to search for position number of i
so input for substr can be dynamic

select substr('vishaljaiswal',instr('vishaljaiswal','i'),6) from dual;

First parameter = vishaljaiswal
2nd parameter : instr('vishaljaiswal','i')
3rd parameter : 6

2nd parameter is decide the starting position , it can be anything for different different strings

MS SP (Position of i) HMC

```
select substr('vishal jaiswal',instr('vishal jaiswal','i'),6) from dual
```

SUBSTR('VISHALJAISWAL',INSTR('VISHALJAISWAL','I'),6)
ishal

Suppose asked to take from 2nd occurrence of "i", give me 6 characters

ms sp (2nd occurrence of i)

select substr('vishal jaiswal',instr('vishal jaiswal','i',1,2),6) from dual

HMC
2nd occurrence

SUBSTR('VISHALJAISWAL',INSTR('VISHALJAISWAL','I',1,2),6)
iswal

ms sp (Position of 'i')

select substr('Abhishek Navle',instr('Abhishek Navle','i'),4) from dual

HMC

SUBSTR('ABHISHEKNAVLE',INSTR('ABHISHEKNAVLE','I'),4)
ishe

→ start from 'i', give me 4 more character

AFTER SPACE → give me 3 more character

ms How many

select substr('ruchi gupta',instr('ruchi gupta','')+1,3) from dual

sp (Position of space)+1 (After space)

SUBSTR('RUCHIGUPTA',INSTR('RUCHIGUPTA','')+1,3)
gup

select substr('aishwarya borker',instr('aishwarya borker','')+1,3) from dual

SUBSTR('AISHWARYABORKER',INSTR('AISHWARYABORKER','')+1,3)
bor

Position of space +1

```
select full_name, substr(FULL_NAME, instr(FULL_NAME, ' ')+1, 3) from co.customers
```

ms 3 character

FULL_NAME	SUBSTR(FULL_NAME, INSTR(FULL_NAME, ' ')+1, 3)
Kris Krill	Kri
Kristin Brooks	Bro
Kristina Livshits	Liv
Kristina Nunez	Nun

} → 3 charc from left name

ms sp (from reverse goto 5th position)

```
select substr('aishwarya borker', -5, 4) from dual
```

4 character

SUBSTR('AISHWARYABORKER', -5, 4)
orke

From the name "Vishal jaiswal" , write a sql to get firstname and lastname

ms sp How many character

```
select substr('vishal jaiswal', 1, instr('vishal jaiswal', ' ')-1) from dual
```

SUBSTR('VISHALJAISWAL', 1, INSTR('VISHALJAISWAL', ' ')-1)
vishal

See the above image, how do I get first name. What is the logic? start from number 1 and stop when i get a space. means start postion is 1, but how many characters means lets find the position of space.

instr('vishal jaiswal', ' ') What this will do. just find the space in "vishal jaiswal" and give you position. but we want to stop before space so do a -1.

Find the last name, just start position is after space.

```
select substr('vishal jaiswal',instr('vishal jaiswal',' ')+1) from dual
```

```
SUBSTR('VISHALJAISWAL',INSTR('VISHALJAISWAL',' ')+1)
```

```
jaiswal
```

'vishalhcst@gmail.com'

--> I want to retrieve my username from email address. Like here. I just want "vishalhcst" as output. it means, I have to start with first position of string and stop when we found "@"

@ position is on number 11. so I have to stop on 10th position.

so, if we do substr('vishalhcst@gmail.com',1,10) --> means start from 1 and stop at 10, it will give us correct output. as vishalhcst

But, the challenge is everyone's userid can be different and the position of @ is also different. It can not be fixed on number 10, so we have to create our query dynamic, so that it will find the position of @ and stop there.

so finding position means, INSTR

if we do INSTR and try to search the position of @, we will get this and this same we have to use in substr.

input: vishalhcst@gmail.com

output: vishalhcst

```
select substr('vishalhcst@gmail.com',1,10) from dual;
```

```
--vishalhcst
```

```
select substr('sroracle7@gmail.com',1,10) from dual;
```

```
--sroracle7@
```

```
select substr('mallakakriti04@gmail.com',1,10) from dual;
```

```
--mallakakri
```

--> So, you can see that 1,10 is not working properly for every email id. because every email id is different. all may have different length user_name. so

we have to find a way to retrieve the username. if we see, "@" is common everywhere. so if we are able to find the position of "@" and stop just before "@", we will get the username.

```
select instr('mallakakriti04@gmail.com','@') from dual; -- 15 --> position of @ is 15
```

```
select instr('sroracle7@gmail.com','@') from dual; -- 10 --> position of @ is 10
```

```
select instr('vishalhcst@gmail.com','@') from dual; -- 11 --> position of @ is 11
```

--> so we can see that position of @ is different. if we use that number, instead of 10, let see

```
select substr('mallakakriti04@gmail.com',1,15) from dual; -- mallakakriti04@ -- if i use 15 then @ is also coming, so let's make it 14
```

```
select substr('mallakakriti04@gmail.com',1,14) from dual; -- mallakakriti04
```

```
select substr('sroracle7@gmail.com',1,10) from dual; -- sroracle7@
```

```
select substr('vishalhcst@gmail.com',1,11) from dual;-- vishalhcst@
```

in SUBSTR there are 3 parameters as we discussed, the 3rd parameter should be dynamic. so in place of 10, we have to write a code, which will search the position of @

```
instr('vishalhcst@gmail.com','@')
```

This will give us the number position of @ and we have to stop before that. so will do minus 1.

```
instr('vishalhcst@gmail.com','@') - 1 will give us the exact number position where we have to stop.
```


sf
Find position of @ & -1 means before that

```
select substr('vishalhcst@gmail.com',1,instr('vishalhcst@gmail.com','@')-1) from dual
```

```
SUBSTR('VISHALHCST@GMAIL.COM',1,INSTR('VISHALHCST@GMAIL.COM','@')-1)
```

```
vishalhcst
```

```
select substr('anandhan.a86@rediffmail.com',1,instr('anandhan.a86@rediffmail.com','@')-1) from dual
```

```
SUBSTR('ANANDHAN.A86@REDIFFMAIL.COM',1,INSTR('ANANDHAN.A86@REDIFFMAIL.COM','@')-1)
```

```
anandhan.a86
```

ms sp How many character

```
select substr('yashsanvatsarkar@yahoo.com',1,instr('yashsanvatsarkar@yahoo.com','@')-1) from dual
```

```
SUBSTR('YASHSANVATSARKAR@YAHOO.COM',1,INSTR('YASHSANVATSARKAR@YAHOO.COM','@')-1)
```

```
yashsanvatsarkar
```

Find the position of @ & -1

ms sp HMC (same find @ & then -1)

```
select substr('mohdfaizan7283@gmail.com',1,instr('mohdfaizan7283@gmail.com','@')-1) from dual
```

```
SUBSTR('MOHDFIAIZAN7283@GMAIL.COM',1,INSTR('MOHDFIAIZAN7283@GMAIL.COM','@')-1)
```

```
mohdfaizan7283
```